

Natural Resources

4.1 WATER RESOURCES

Water is an indispensable natural resource. Life on the earth could originate because of the presence of water. About 97% of earth's surface is covered by water. Water is essential for maintenance of life. Most of the animals and plants have 60–65% water in their bodies. Due to its unique properties water is of multiple uses for all living organisms. Water is absolutely essential for life. Most of the life processes take place in water contained in the body. Uptake of nutrients, their distribution in the body, regulation of temperature, and removal of wastes are all mediated through water. Human beings depend on water for almost every developmental activity. Water is used for drinking, irrigation, transportation, washing and waste disposal for industries and used as a coolant for thermal power plants. Water shapes the earth's surface and regulates our climate.

4.2 AVAILABILITY AND QUALITY ASPECTS

4.2.1 Water Availability

Although water is abundant on this earth, yet it is very precious. Out of the total water reserves of the world, about 97.4% is salty water (marine) and only 2.6% is fresh water. Even this small fraction of fresh water is not available to us as most of it is locked up in polar ice caps (1.98%) and just 0.6% is readily available to us in the form of groundwater and surface water (rivers and lakes), water in atmosphere and in the living organisms.

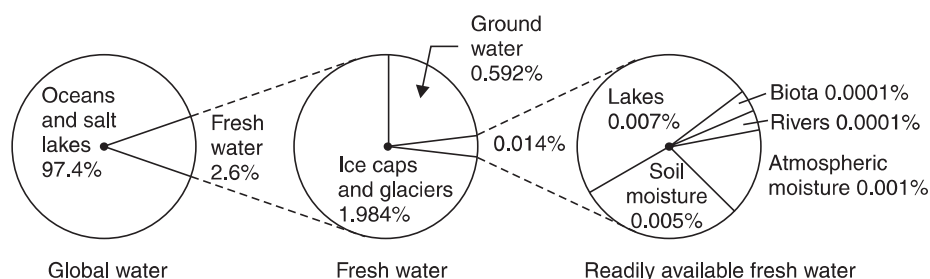


Fig. 4.1. Distribution of global water.

Oceans: Oceans are ultimate sinks for all types of water. About 97.4% of total water is present in oceans. Salt content of oceans is about 3.5%. Sea water is not fit for human consumption and other anthropogenic activities without desalination.

Glaciers and Ice: A major part of available approximately 2.6% of fresh water is locked up into glaciers and ice caps. These occur at high altitude or high latitude. Antarctic glacier contains approximately 85% of all the world's ice. About 10% is made by greenland ice-sheet and arctic ice. Rest 5% is in the form of snow on mountain peaks.

Groundwater

About 0.59% of the total water resources is in the form of groundwater and it is about 35–50 times that of surface water supplies. Till some time back groundwater was considered to be very pure. However, of late, even groundwater aquifers have been found to be contaminated by leachates from sanitary landfills etc.

A layer of sediment or rock that is highly permeable and contains water is called an **aquifer**. Layers of sand and gravel are good aquifers while clay and crystalline rocks (like granite) are not since they have low permeability. Aquifers may be of two types:

Unconfined aquifers which are overlaid by permeable earth materials and they are recharged by water seeping down from above in the form of rainfall and snow melt.

Confined aquifers which are sandwiched between two impermeable layers of rock or sediments and are recharged only in those areas where the aquifer intersects the land surface. Sometimes the recharged area is hundreds of kilometers away from the location of the well. Figure 4.2 shows the groundwater system. Groundwater is not static, it moves, though at a very slow rate of about a meter or so in a year.

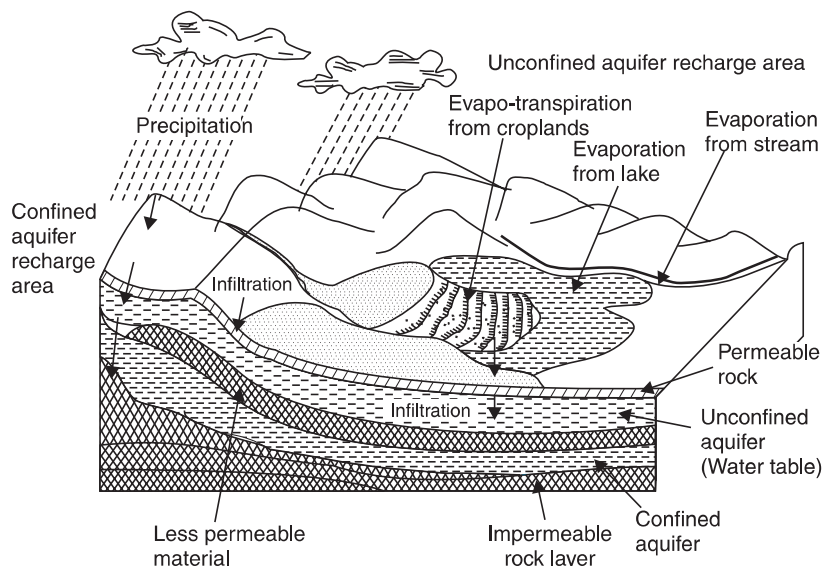


Fig. 4.2 The groundwater system. An unconfined aquifer (water table) is formed when water collects over a rock or compact clay. A confined aquifer is formed as sandwiched between two layers having very low permeability.

Effects of Groundwater Usage

- (i) **Subsidence:** When groundwater withdrawal is more than its recharge, the sediments in the aquifer get compacted, a phenomenon known as *ground subsidence*. Huge economic losses may occur due to this phenomenon because it results in the sinking of overlying land surface. The common problems associated with it include structural damage in buildings, fracture in pipes, reversing the flow of sewers and canals and tidal flooding.
- (ii) **Lowering of water table:** Mining of groundwater is done extensively in arid and semi-arid regions for irrigating crop fields. However, it is not advisable to do excessive mining as it would cause a sharp decline in future agricultural production, due to lowering of water table.
- (iii) **Water logging:** When excessive irrigation is done with brackish water it raises the water table gradually leading to water-logging and salinity problems.

Surface Water

The water coming through precipitation (rainfall, snow) when does not percolate down into the ground or does not return to the atmosphere as evaporation or transpiration loss, assumes the form of streams, lakes, ponds, wetlands or artificial reservoirs known as surface water. The surface water is largely used for irrigation, industrial use, public water supply, navigation etc. A country's economy is largely dependent upon its rivers.

Water rich vs. Water poor countries

The top ten water rich countries are Iceland, Surinam, Guyana, Papua New Guinea, Gabon, Solomon Islands, Canada, Norway, Panama, and Brazil lying in the far north and have low evaporation losses.

The water poor countries include Kuwait, Egypt, United Arab Emirates, Malta, Jordan, Saudi Arabia, Singapore, Maldives, Israel and Oman lying in the desert belt at about 15° to 25° latitude and some of them like Malta and Singapore are densely populated areas resulting in low per capita water.

Water in the rivers comes from precipitation, melting of snow and ice and groundwater seepage.

Lakes and ponds are shallow depressions in the earth. Some lakes are a few km in length. These are permanent or temporary water bodies. These suffer from the problem of siltation which may make them temporary in nature. Lakes and ponds contain more than 100 times the water present in all rivers and streams.

Swamps and marshes are important wetlands. These wetlands have the capacity to absorb surface run-off. These provide sufficient time for recharge of groundwater and also play an important role in hydrological cycle.

Soil moisture is held up in the soil capillaries and this water is useful for plant growth. **Atmosphere** contains about 0.001% of the total water on this earth. It forms a small proportion of total water available. Atmospheric water stays in the atmosphere for short duration as it gets distributed with air currents and falls as precipitation to replenish the terrestrial water bodies.

Hydrological Cycle:

The water we use keeps on cycling endlessly through the environment which we call as **Hydrological Cycle**. We have enormous resources of water on the earth amounting to about 1404 million km³. The water from various moist surfaces evaporates and falls again on the earth in the form of rain or snow and passes through living organisms and ultimately returns to the oceans. Every year about 1.4 inch thick layer of water evaporates from the oceans, more than 90% of which returns to the oceans through the hydrological cycle. Solar energy drives the water cycle by evaporating it from various water bodies which subsequently return through rainfall or snow. Plants too play a very important role by absorbing the groundwater from soil and releasing it into the atmosphere by the process of transpiration.

Global distribution of water resources is quite uneven depending upon several geographic factors. Tropical rain forest areas receive maximum rainfall while the major world deserts occur in zones of dry, descending air (20–40° N and S) and receive very little rainfall.

4.2.2 Water Quality

Water is characterized by certain unique qualities which make it a marvellous resource:

- (i) It exists as a liquid over a wide range of temperature *i.e.* from 0° to 100°C.
- (ii) It has the highest specific heat, due to which it warms up and cools down very slowly without causing shocks of temperature jerks to the aquatic life.
- (iii) It has a high latent heat of vaporization. Hence, it takes a huge amount of energy for getting vaporized. That's why it produces a cooling effect as it evaporates.
- (iv) It is an excellent solvent for several nutrients. Thus, it can serve as a very good carrier of nutrients, including oxygen, which are essential for life. But, it can also easily dissolve various pollutants and also becomes a carrier of pathogenic microorganisms.
- (v) Due to high surface tension and cohesion it can easily rise to great heights through the trunk even in the tallest of the trees like *Sequoia*.
- (vi) It has an anomalous expansion behaviour *i.e.* as it freezes, it expands instead of contracting and thus becomes lighter. It is because of this property that even in extreme cold, the lakes freeze only on the surface. Being lighter the ice keeps floating, whereas the bottom waters remain at a higher temperature and therefore, can sustain aquatic organisms even in extreme cold.

As water is a good solvent, it is prone to contamination by a variety of substances. Water bodies have been treated as sewers for industrial and municipal discharges. Deterioration of fresh water renders it unfit for drinking. Water treatment is done before domestic supply.

More than 90% of the population in the United State as also many European countries don't even bother whether the drinking water is safe or not. However, in the developing countries the first question a visitor asks is whether the drinking water is safe to drink. We quite often see most of such people carrying their own mineral water bottles. It is estimated that 80 per cent of sickness in the world is due to improper quantity or quality of drinking water. The most affected areas in the world are in Asia, Africa and Latin America. About 60% of babies born in the developing nations die of gastric disturbances. Millions of people are affected by schistosomiasis and filariasis. It is imperative to provide drinking water which is free from pathogens and other noxious substances which can cause sickness or disease.

Drinking water should

- (i) be free from pathogens
- (ii) have required salts and should not have high concentration of minerals and organic matter.
- (iii) not have toxic chemicals

Drinking Water Standards. The drinking water standards are the values for each parameter beyond which water is considered unfit for drinking. These standards are given in table 4.1

Table 4.1. Drinking Water – Specification (BIS 10500: 1991)

Sl. No.	Substance or characteristic	Requirement (Desirable limit)	Permissible limit in the absence of alternate source
Essential Characteristics			
1.	Colour (Hazen units, Max)	5	25
2.	Odour	Unobjectionable	Unobjectionable
3.	Taste	Agreeable	Agreeable
4.	Turbidity (NTU. Max)	5	10
5.	pH Value	6.5 to 8.5	No relaxation
6.	Total hardness (as CaCO ₃) mg/L, Max	300	600
7.	Iron (as Fe) mg/L, Max	0.3	1.0
8.	Chlorides (as Cl) mg/L, Max.	250	1000
9.	Residual free chlorine, mg/L, Min.	0.2	--
10.	Fluoride (as F) mg/L, Max	1.0	1.5
Desirable Characteristics			
11.	Dissolved solids mg/L, Max	500	2000
12.	Calcium (as Ca) mg/L, Max	75	200
13.	Magnesium (as Mg) mg/L, Max	30	100
14.	Copper (as Cu) mg/L, Max	0.05	1.5
15.	Manganese (as Mn) mg/L, Max	0.10	0.3
16.	Sulfate (as SO ₄) mg/L, Max	200	400
17.	Nitrate (as NO ₃) mg/L, Max	45	No relaxation
18.	Phenolic compounds (as C ₆ H ₅ OH) mg/L, Max	0.001	0.002
19.	Mercury (as Hg) mg/L, Max	0.001	No relaxation

20.	Cadmium (as Cd) mg/L, Max	0.01	No relaxation
21.	Selenium (as Se) mg/L, Max	0.01	No relaxation
22.	Arsenic (as As) mg/L, Max	0.01	No relaxation
23.	Cyanide (as CN) mg/L, Max	0.05	No relaxation
24.	Lead (as Pb) mg/L, Max	0.05	No relaxation
25.	Zinc (as Zn) mg/L, Max	5	15
26.	Anionic detergents (as MBAS) mg/L, Max	0.2	1.0
27.	Chromium (as Cr ⁶⁺) mg/L, Max	0.05	No relaxation
28.	Polynuclear aromatic hydrocarbons (as PAH) g/L, Max	--	--
29.	Mineral oil mg/L, Max	0.01	0.03
30.	Pesticides mg/L, Max	Absent	0.001
31.	Radioactive materials		
	(i) Alpha emitters Bq/L. Max	--	0.1
	(ii) Beta emitters pci/L. Max	--	1.0
32.	Alkalinity mg/L, Max	200	600
33.	Aluminium (as Al) mg/L, Max	0.03	0.2
34.	Boron mg/L, Max	1	5

Bacteriological Examination

Water in Distribution System

Water in the distribution system, piped water, upto the consumer's level should have no coliform organisms. Since it is not ideally possible, therefore, the following standard in the water sample collected from distribution system is recommended (tested in accordance with IS 1622:1981)

- (a) Throughout any year, 95 percent of samples should not contain any coliform organisms in 100 mL;
- (b) No sample should contain *E. Coli* in 100 mL;
- (c) No sample should contain more than 10 coliform organisms per 100 mL; and
- (d) Coliform organisms should not be detectable in 100 mL of any two consecutive samples.

4.3 WATER BORNE AND WATER INDUCED DISEASES

Disease causing agents (pathogens) are infectious organisms which cause disease when these grow and multiply in the host. These organisms come from the waste of infected individuals which reach the municipal wastewater. Municipal water usually contains various types of bacteria, viruses, protozoans, worms and other types of agents which cause various types of diseases of man and other animals. Contaminated water can cause water borne and water-induced and water-contact diseases.

Water borne diseases are caused by ingestion of pathogens. These pathogens can be ingested with contaminated drinking water, hands or utensils. Some of the waterborne diseases caused by bacteria or virus are cholera, bacterial dysentery, typhoid, polio, infectious hepatitis. Certain protozoans also cause diseases like amoebic dysentery, cryptosporidiosis, Giardiasis etc. Giardiasis is caused by **Giardia lamblia**. The cysts of **Giardia** pass through the faeces of carrier humans/animals and can survive for many months in the environment.

Cholera is caused by a bacterium (**Vibrio cholerae**) and causes severe diarrhoea, vomiting which may cause cramps. Bacterial dysentery is caused by a bacterium (**Shigella dysenteriae**). The patient suffers from abdominal pain (due to infection of colon) and diarrhoea with mucus and blood in stools.

Enteritis is caused by bacterium (*Clostridium perfringens*). The small intestine develops inflammation which results in loss of appetite, abdominal cramps and diarrhoea. Typhoid is also a bacterial disease caused by (*Salmonella typhi*). The symptoms include headache, weakness, fever, rashes and sometime bleeding in intestines.

Infectious hepatitis is caused by virus (Hepatitis virus A). In this disease jaundice is caused due to liver inflammation. Other symptoms include headache, nausea, vomiting and loss of appetite.

Poliomyelitis is caused by virus (Poliovirus). The disease is characterised by sore throat, fever, diarrhoea, pain in limbs and back. Paralysis and atrophy of muscles can occur if infection spreads to spinal cord.

Cryptosporidiosis is caused by protozoan (*Cryptosporidium* sp.) The disease symptoms are diarrhoea and cramps. Symptoms last for up to 22 days.

Amoebic dysentery is caused by a protozoan (**Entamoeba histolytica**). Symptoms of the disease are abdominal pain. Due to infection of colon diarrhoea becomes painful. Mucus and blood are seen in stool.

WHO Report: 13 million deaths due to environmental exposure

The World Health Organisation in its report (2006) has emphasised that diseases can be prevented through healthy environment. More than 33 per cent of diseases in children below 5 yrs. are due to environmental exposures. About one quarter of diseases in the world are due to environmental exposures while it is one third in less developed countries.

Malaria and diarrhoea are the world's two biggest killer diseases of childhood. Other such diseases are due to infection of respiratory tract. These diseases can be prevented through better environmental management.

Safe storage of water, better hygienic environment, use of safer and cleaner fuels, increased safety measures at home and workplace and reduction and proper management of wastes and hazardous substances can reduce the environmental disease burden.

It is estimated that 94 per cent of diarrhoeal burden of disease comes from unsafe water, sanitation and hygiene and 41 per cent of lower respiratory infections come from indoor and outdoor air pollution.

There are some diseases which may occur without ingestion of pathogens with drinking water. These are called water-contact diseases and are spread by entry into the host of the parasite or its developmental stage from the contaminated water, *e.g.*, Schistosomiasis, Ancylostomiasis, Dracunculiasis.

There are other water related diseases which are caused by insect vector like mosquitoes which breed in the contaminated water *e.g.* dengue, filariasis, yellow-fever, malaria, onchocerciasis, trypanosomiasis etc. Some diseases are caused due to poor personal hygiene and skin and eye contact with contaminated water *e.g.* scabies, trachoma and other lice, tick and flea borne diseases.

Some diseases are induced by contaminated water. These water-induced diseases are due to the presence of arsenic, fluoride, nitrates from fertilizers, carcinogenic pesticides, heavy metals etc. The details of such diseases are given in water pollution chapter of unit III.

About 11 crore (1.1 billion) people, of which about 91% from Asia and Africa don't have access to good quality water. Contaminated water is responsible for about 80% of the infectious diseases. More than 50 lakh people die each year due to such diseases and more than 20 lakh people, most of them small children, die each year from diarrhoea caused by contaminated water.

Such diseases caused by contaminated water can be minimised or controlled by the following approach:

- Open defecation should be avoided.
- Diapers should be disposed off properly.
- Hands should be washed before eating.
- Raw vegetables and fruits should be washed properly before eating.
- General sanitation be improved.
- Public health department should properly monitor and any outbreak of water borne diseases should be duly reported to concerned department.
- Media can help in disseminating information and status to general public.
- Education can spread awareness among people of different ages.

4.4 FLUORIDE PROBLEM IN DRINKING WATER

Fluoride is present in varying levels in almost all types of waters. Sea water contains about 1 mg/L, rivers and lakes less than 0.5 mg/L and ground water may have from low to high values depending upon nature of rock and fluoride bearing minerals in the bedrock. Fluoride may enter in the public water system from natural source *i.e.*, run-off from weathering of fluoride containing rocks and soils or leaching from soil to ground water. Industrial discharges may also contaminate water supply.

Requirement of fluoride: Some amount of fluoride is required in drinking water to prevent tooth decay. Therefore, fluoride is intentionally added to public drinking water supply where natural fluoride is low. Optimal level which can prevent tooth decay is from 0.7 to 1.2 mg/L.

Factors affecting intake of fluoride: Fluoride intake per person per day depends on fluoride concentration in drinking water and the amount of water taken by an individual per day. Water consumption increases with temperature, humidity, exercise, state of health and other factors including diet. Climatic conditions influence water intake by an individual. It is more near equator.

As per a rough estimate total daily fluoride intake in temperate climate would be approximately 0.6 mg per adult per day where no fluoride is added to drinking water and 2 mg per adult per day in a fluoridated area.

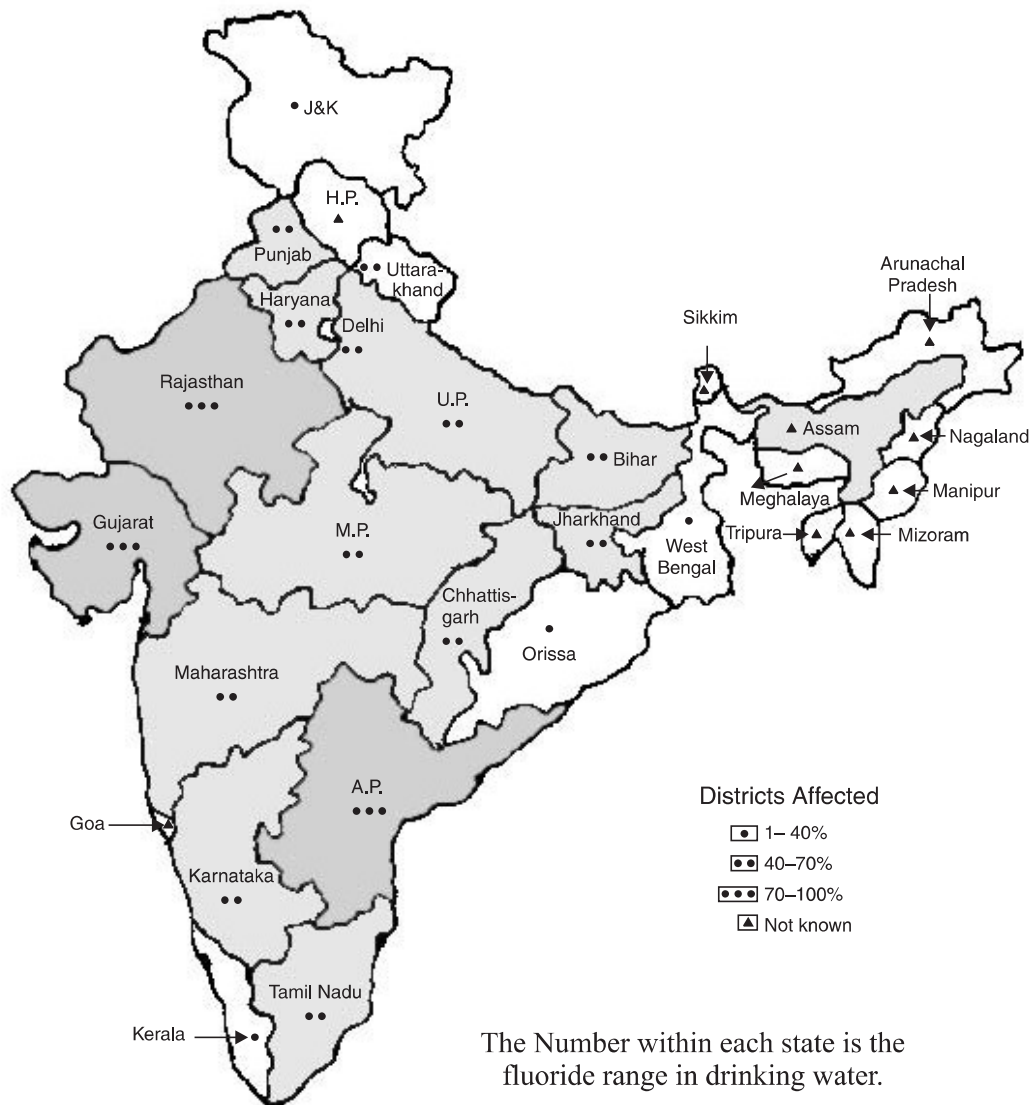
Sources of fluoride in ground water: Fluoride problem may be endemic *i.e.*, restricted to the areas with high fluoride levels. However, the problem is prevalent in various geographical areas of the world. The occurrence of high levels are observed in the areas with:

- (i) Sediments of marine origin in mountainous areas *e.g.*, Syria, Turkey, Mediterranean regions, Iraq, Iran etc.
- (ii) Volcanic rocks-Jordan, Sudan, Uganda, Kenya, Ethiopia, Nairobi, Tanzania, etc. Soda lakes have high fluoride concentration *i.e.*, upto 2800 mg/L.
- (iii) Igneous and metamorphic rocks. *e.g.*, Granitic and gneissic rocks. *e.g.*, India, Pakistan, China, Thailand, Sri Lanka, southern Africa, west Africa.

Fluoride in ground water comes due to dissolution of fluorite, apatite and topaz from local bedrock. Presence of calcium limits fluoride concentration in water. High concentration of fluoride has been observed in aquifers poor in calcium and rich in fluoride bearing minerals.

Problem of fluorosis in India: In India fluorosis is a serious national problem. The seriously affected areas are villages in Andhra Pradesh, Punjab, Haryana, Rajasthan, Uttar Pradesh, Tamil Nadu and Gujarat.

The highest concentration, to date, has been reported from Rewari district of Haryana *i.e.*, 48 mg/L. An estimated 62 million people including 6 million children in India are affected by various types of fluorosis.



Source: UNICEF State of Art Report, 1999

Fig. 4.3. Fluorosis prevalent states of India.

Although it is difficult to assess exact extent of affected people as the monitoring of ground water of wells and handpumps is not uniform *i.e.*, all villages of the district are not monitored or samples are taken from some handpumps in a particular locality ignoring others. As per WHO estimates about 10 to 25% of rural population of various states is at risk.

Health problems due to fluorosis: Fluoride exposure may lead to dental (enamel) fluorosis and skeletal fluorosis (Fig. 4.4). Severity of fluorosis depends on fluoride exposure and period of development during exposure. Dental fluorosis may vary from mild discoloration of tooth surface to severe yellow or brown striations or mottling, loss of enamel and dental cavities. In the U.S. prevalence of severe enamel

fluorosis is very low below 2 mg/L of fluoride in drinking water. High fluoride exposure in terms of high levels in drinking water and high duration of exposure may lead to crippling skeletal fluorosis characterised by osteosclerosis, calcification of ligaments and tendons and extreme bone deformities.

Carcinogenicity of fluoride in the tests on experimental animals could not be established. No effects have been observed on reproduction, reproductive organs or foetus development.

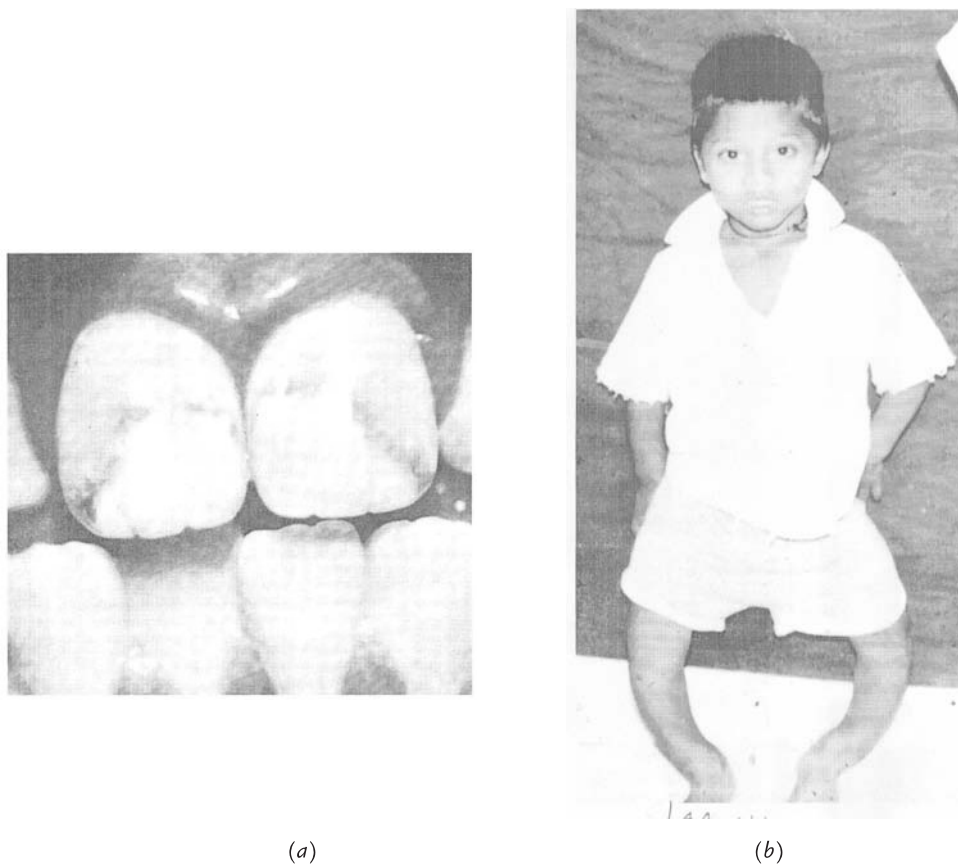


Fig. 4.4. Dental fluorosis (a) and skeletal fluorosis (b).

Fluoride standards in drinking water: WHO has set the value of fluoride in drinking water at 1.5 mg/L. However, this guideline value of WHO is not a fixed one but is supposed to be adopted in consideration with the local conditions.

Fluoride removal from drinking water: Where alternative source of drinking water is not available, defluoridation of drinking water is the only remedy. It can be done with the help of bone charcoal. **Nalgonda process** developed and adopted by National Environmental Engineering Institute (NEERI) can be used at community and household level. The process uses aluminium sulfate to remove fluoride. The flocs of aluminium hydroxide formed in the process takes a few hours to settle which may be discarded.

4.5 MINERAL RESOURCES

Minerals are naturally occurring, inorganic, crystalline solids having a definite chemical composition and characteristic physical properties. There are thousands of minerals occurring in different parts of the world. However, most of the rocks, we see everyday are just composed of a few common minerals like quartz, feldspar, biotite, dolomite, calcite, laterite etc. These minerals, in turn, are composed of some elements like silicon, oxygen, iron, magnesium, calcium, aluminium etc.

Uses

Minerals find use in a large number of ways in everyday use in domestic, agricultural, industrial and commercial sectors and thus form a very important part of any nation's economy. The main uses of minerals are as follows:

- (i) Development of industrial plants and machinery. *e.g.* iron, coal, aluminium, lead chromium, nickel, mercury, cadmium etc.
- (ii) Generation of energy *e.g.* coal, lignite, uranium.
- (iii) Construction, housing, settlements, *e.g.* iron, aluminium, silicate, limestone.
- (iv) Defence equipments (weapons, armaments) *e.g.* copper, chromium, cobalt, manganese, iron, lead.
- (v) Transportation means *e.g.* iron, lead, aluminium, platinum.
- (vi) Communication (telephone wires, cables, electronic devices) *e.g.* copper, lead, nickel.
- (vii) Medicinal system (particularly in Ayurvedic system) *e.g.* gold, silver, iron.
- (viii) Formation of alloys for various purposes *e.g.* steel alloys.
- (ix) Agriculture (as fertilizers, seed dressings and fungicides) *e.g.* zineb containing zinc, maneb containing manganese etc.
- (x) Jewellery *e.g.* gold, silver, platinum, diamond.

Based on their properties, minerals are basically of two types:

- (i) Non-metallic minerals *e.g.* graphite, diamond, quartz, feldspar.
- (ii) Metallic minerals *e.g.* bauxite, laterite, haematite etc.

Use of metals by human beings has been so extensive since the very beginning of human civilization that two of the major epochs of human history are named after them as Bronze Age and Iron Age. The reserves of metals and the technical know-how to extract them have been the key elements in determining the economy and political power of nations. Out of the various metals, those used in maximum quantity are iron and steel (740 million metric tons annually) followed by manganese, copper, chromium, aluminium and nickel.

Distribution and uses of some of the major metallic and non-metallic minerals are given in Tables 4.2 and 4.3.

**Table 4.2. Major Reserves and Important Uses
of Some of the Major Metals**

<i>Metal</i>	<i>Major world reserves</i>	<i>Major uses</i>
Aluminium	Australia, Guinea, Jamaica	Packaging food items, transportation, utensils, electronics
Chromium	CIS, South Africa	For making high strength steel alloys, in textile/tanning industries
Copper	U.S.A., Canada, CIS, Chile, Zambia	Electric and electronic goods, building, construction, vessels
Iron	CIS, South America, Canada, U.S.A.	Heavy machinery, steel production transportation means
Lead	North America, U.S.A., CIS	Leaded gasoline, Car batteries, paints, ammunition
Manganese	South Africa, CIS, Brazil, Gabon	For making high strength, heat-resistant steel alloys
Platinum group	South Africa, CIS	Use in automobiles, catalytic converters, electronics, medical uses.
Gold	South Africa, CIS, Canada	Ornaments, medical use, electronic use, use in aerospace
Silver	Canada, South Africa, Mexico	Photography, electronics, jewellery
Nickel	CIS, Canada, New Caledonia	Electroplating

Table 4.3. Major Uses of Some Non-metallic Minerals

<i>Non-metal mineral</i>	<i>Major uses</i>
Silicate minerals	Sand and gravel for construction, bricks, paving etc.
Limestone	Used for concrete, building stone, used in agriculture for neutralizing acid soils, used in cement industry
Gypsum	Used in plaster wall-board, in agriculture
Potash, phosphorite	Used as fertilizers
Sulphur pyrites	Used in medicine, car battery, industry.

It is evident from the Tables that the CIS countries (The Commonwealth of Independent States *i.e.* 12 republics of former USSR), the United States of America, Canada, South Africa and Australia are having the major world reserves of most of the metallic minerals. Due to huge mineral and energy resources, the USA became the richest and the most powerful nation in the world in even less than 200 years. Japan too needs a mention here, as there are virtually no metal reserves, coal, oil and timber resources in Japan and it is totally dependent on other countries for its resources. But, it has developed energy efficient technologies to upgrade these resources to high quality finished products to sustain its economy.

Minerals are sometimes classified as **Critical** and **Strategic**.

Critical minerals are essential for the economy of a nation *e.g.* iron, aluminium, copper, gold etc.

Strategic minerals are those required for the defence of a country *e.g.* Manganese, cobalt, platinum, chromium etc.

Some Major Minerals of India

(a) Energy generating minerals

Coal and lignite: West Bengal, Jharkhand, Orissa, Madhya Pradesh, Andhra Pradesh.

Uranium (Pitchblende or Uranite ore): Jharkhand, Andhra Pradesh (Nellore, Nalgonda), Meghalaya, Rajasthan (Ajmer).

(b) Other commercially used minerals

Aluminium (*Bauxite ore*): Jharkhand, West Bengal, Maharashtra, Madhya Pradesh, Tamil Nadu.

Iron (*Haematite and magnetite ore*): Jharkhand, Orissa, Madhya Pradesh, Andhra Pradesh, Tamil Nadu, Karnataka, Maharashtra and Goa.

Copper (*Copper pyrites*): Rajasthan (Khetri), Bihar, Jharkhand, Karnataka, Madhya Pradesh, West Bengal, Andhra Pradesh and Uttarakhand.

4.6 FOREST WEALTH

Forests are one of the most important natural resources on this earth. Covering the earth like a green blanket these forests not only produce innumerable material goods, but also provide several environmental services which are essential for life.

Forests are of immense value to us. They are not only useful for industry but also for rural economic growth. They offer huge potential for reducing poverty while also conserving their valuable key resources.

Figure 4.5 compares the value of forests in terms of economic products and environmental services.

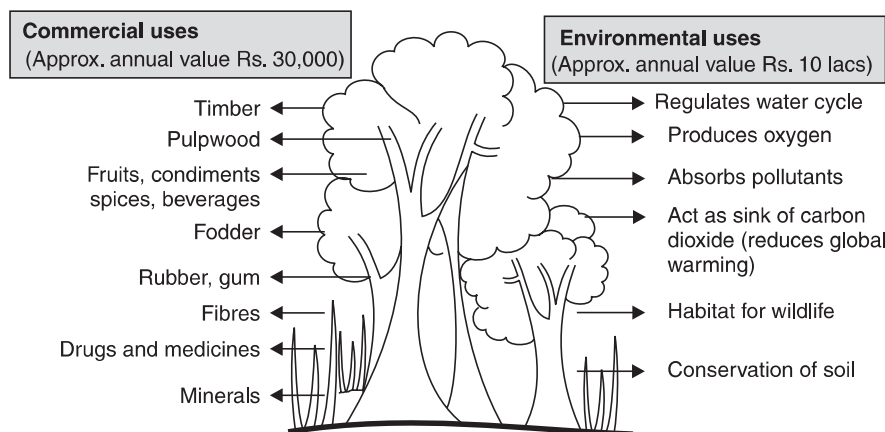


Fig. 4.5. Economic vs. Environmental value of forests.

Economic uses: Forests provide us a large number of commercial goods which include timber, firewood, pulpwood, food items, gum, resins, non-edible oils, rubber, fibres, lac, bamboo canes, fodder, medicine, drugs and many more items.

Half of the timber cut each year is used as fuel for heating and cooking. One third of the wood harvest is used for building materials as lumber, plywood and hardwood, particle board and chipboard. One sixth of the wood harvest is converted into pulp and used in paper industry. Many forest lands are used for mining, agriculture, grazing, and recreation and for development of dams.

Ecological uses: The ecological services provided by our forests may be summed up as follows:

- **Production of oxygen:** The trees produce oxygen by photosynthesis which is vital for life on this earth. They are rightly called as earth's lungs.
- **Reducing global warming:** The main greenhouse gas carbon dioxide (CO_2) is absorbed by the forests as a raw material for photosynthesis. Thus forest canopy acts as a sink for CO_2 thereby reducing the problem of global warming caused by greenhouse gas CO_2 .
- **Wildlife habitat:** Forests are the homes of millions of wild animals and plants. About 7 million species are found in the tropical forests alone.
- **Regulation of hydrological cycle:** Forested watersheds act like giant sponges, absorbing the rainfall, slowing down the runoff and slowly releasing the water for recharge of springs. About 50–80 % of the moisture in the air above tropical forests comes from their transpiration which helps in bringing rains.
- **Soil conservation:** Forests bind the soil particles tightly in their roots and prevent soil erosion. They also act as wind-breaks.
- **Pollution moderators:** Forests can absorb many toxic gases and can help in keeping the air pure. They have also been reported to absorb noise and thus help in preventing air and noise pollution.

While a typical tree produces annually commercial goods worth about Rs. 30,000 it provides environmental services worth nearly Rs. one lakh.

4.7 MATERIAL CYCLES

Various materials including different nutrients and metals move in the ecosystem in a cyclic manner. The major reserves or storage compartment of the materials are known as reservoirs.

When the major reservoir of a nutrient is in the atmosphere, it is known as a **gaseous cycle**. *e.g.*, nitrogen cycle, which has its reservoir in the form of nitrogen gas (N_2) constituting about 78% of the atmosphere. When the reservoir is in the earth's crust or sediments, it is known as a **sedimentary cycle** *e.g.*, phosphorus cycle—which has its reserve as phosphate rocks. Sulphur cycle is an example of an intermediate type, which has reservoir both in soil and the atmosphere.

The amount of material and its rate of movement in a cycle is referred to as **flux**, generally expressed as g/yr or a similar expression. Movement of the materials from one reservoir to another may be driven by physical agents like wind or gravitational energy. It may also be due to chemical energy, *e.g.*, when the water body reaches saturation—the reservoir is chemically full and therefore, no longer can hold it as such. Then the material usually is precipitated out.

The average time for which a material (molecule of a substance) remains in a reservoir is known as its **residence time**. *e.g.*, residence time of heavy metals like mercury and lead is quite long in animal bodies but the residence time of lead (Pb) in the atmosphere is quite short. (about four days).

Nutrients like carbon, nitrogen, sulphur, oxygen, hydrogen, phosphorus etc. move in circular paths through biotic and abiotic components and are known as **biogeochemical cycles**. Water also moves in a cycle, known as hydrological cycle. The nutrients too move through the food chain and ultimately reach the detritus compartment (containing dead organic matter) where various microorganisms carry out decomposition. Various organically bound nutrients of dead plants and animals are converted into inorganic substances by microbial decomposition that are readily used up by plants (primary producers) and the cycle starts afresh.

4.7.1 Carbon Cycle

Sometimes human interferences disturb the normal cycling of nutrients and create ecosystem imbalance. For example, nature has a very balanced carbon cycle (Fig. 4.6). Carbon, in the form of carbon dioxide is taken up by green plants as a raw material for photosynthesis, through which a variety of carbohydrates and other organic substances are produced. It moves through the food chain and ultimately organic carbon present in the dead matter is returned to the atmosphere as carbon dioxide by microorganisms. Respiration by all organisms produces carbon dioxide, while the latter is used up by plants.

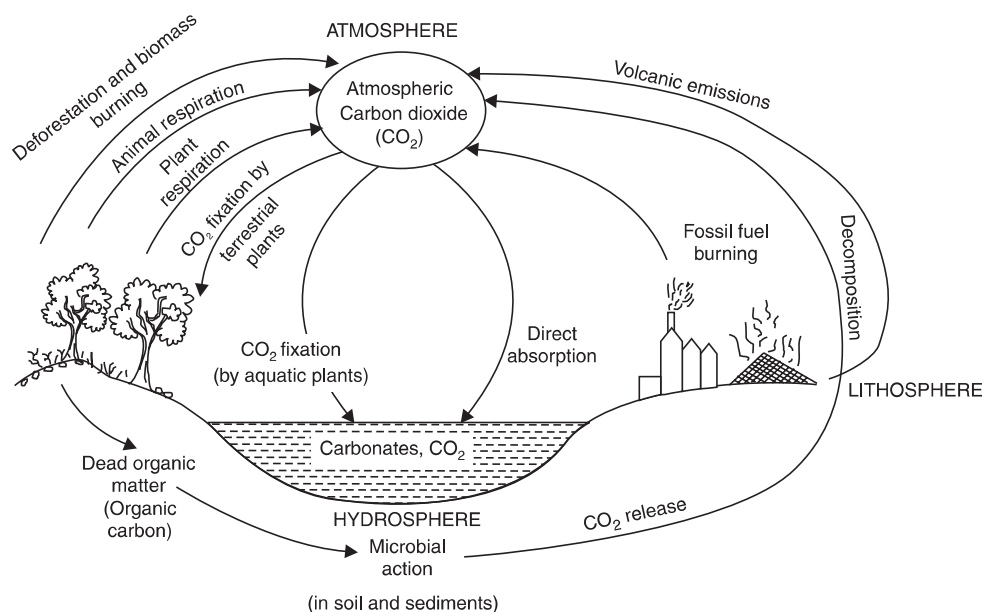


Fig. 4.6. Carbon cycle, showing both natural and man-made fluxes.

Carbon plays a crucial role in regulation of global climate. Human activities have caused carbon fluxes to increase by releasing carbon as carbondioxide from reservoirs through burning of fossil fuels and plant biomass. Deforestation also plays an important role as the sink of CO₂ is lost due to loss of tree canopy. The forest canopies (leaf cover) can otherwise act as excellent CO₂ sinks by using the atmospheric CO₂ for photosynthesis. Thus, there has been an increase in the atmospheric carbon over the past two hundred years, that is resulting in global warming problem.

4.7.2 Nitrogen Cycle

Nitrogen is present in the atmosphere as N₂ in large amount (78%) and it is fixed either by the physical process of lightening or biologically by some bacteria and/or cyanobacteria (blue green algae). The nitrogen is taken up by plants and used in metabolism for biosynthesis of amino acids, proteins, vitamins etc. and passes through the food chain. After death of the plants and animals, the organic nitrogen in dead tissues is decomposed by several groups of ammonifying bacteria converting organic -N into ammonia and nitrifying bacteria like *Nitrosomonas* and *Nitrobacter* which convert them into nitrites and nitrates respectively, which are again used by plants. Some bacteria like *Pseudomonas* convert nitrates, into molecular nitrogen or N₂ by **denitrification** which is released back into the atmosphere and the cycle goes on.

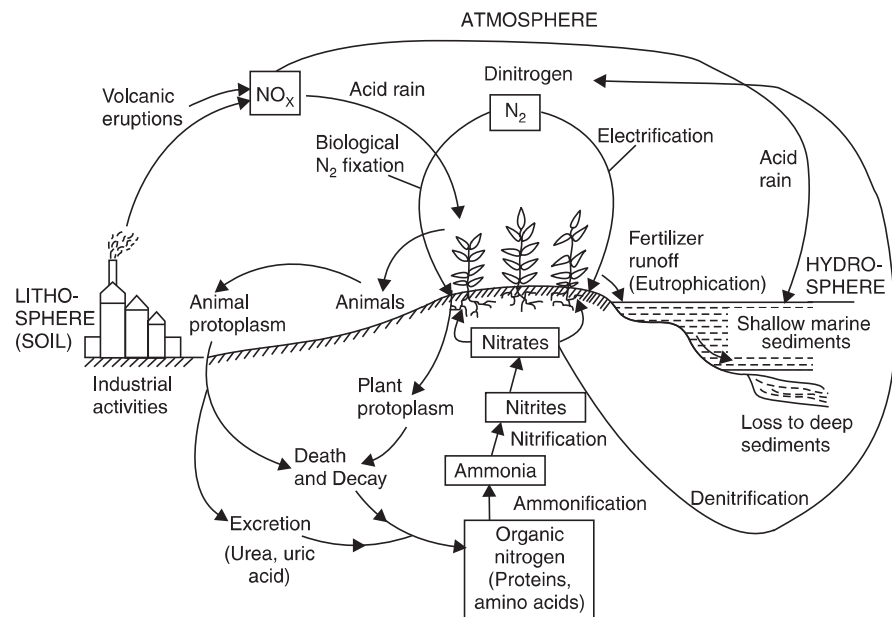


Fig. 4.7. Nitrogen cycle—a gaseous cycle with major reserve as N_2 (78%) in the atmosphere. Circulation of N- between living components and soil/atmosphere is mediated by a group of micro-organisms which convert one form of N into another.

4.7.3 Sulphur Cycle

Sulphur has its reservoir both in atmosphere (as oxides of sulphur, hydrogen sulphide) and in rocks and sediments (as mineral pyrites). Atmospheric sulphur plays an important role, as the oxides of sulphur

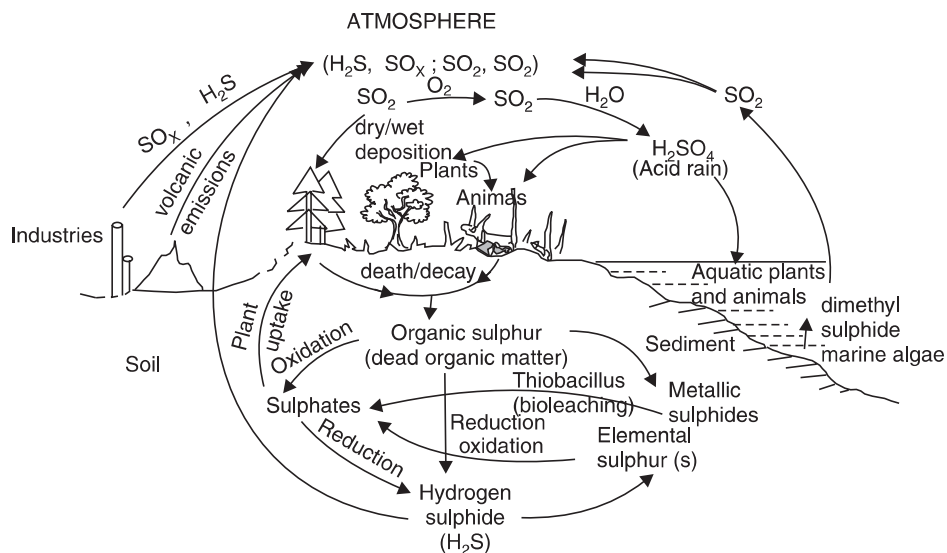


Fig. 4.8. Sulphur cycle—an intermediate type cycle with reservoir of S both in atmosphere and sediments. In the sediments transformations take place from one form to another through microbe mediated oxidation–reduction reactions. Human activities have altered the fluxes of S-cycle mainly by fossil fuel burning.

react with rain water producing sulphuric acid, which comes down as 'acid rain'. Oxides of sulphur (SO_x) are released into the atmosphere due to burning of fossil fuels. Thus human beings play a significant role in the material cycling of sulphur. In the soil or water, there are different groups of micro-organisms which carry out oxidation and reduction of various sulphur compounds. A special role is played by **Thiobacillus** bacterium, which converts sulphides into sulphuric acid. These bacteria help in bio-leaching of metals from ores containing pyrites (S) as impurities. Figure 4.8 illustrates the complex sulphur cycle.

I. QUESTIONS

1. Discuss the importance and availability of water on the earth.
2. What is an aquifer? Discuss its types.
3. Discuss hydrological cycle and its importance.
4. What are water borne diseases? Name a few of them.
5. What are the sources of fluoride in ground water?
6. What are the uses of various types of minerals?
7. Discuss the major environmental impacts of mineral extraction.
8. Discuss major uses of forests. How would you justify that ecological uses of forests surpass commercial uses?
9. What are major causes and consequences of deforestation?
10. Describe carbon cycle. How does carbon in the environment play a role in regulating global climate?

II. OBJECTIVE TYPE QUESTIONS

(A) FILL IN THE BLANKS

1. About % of the earth's surface is covered by water.
2. Only % of total water on earth is readily available to us in the form of groundwater and fresh water.
3. A layer of sediment or rock that is highly permeable and contains water (ground water) is called an
4. Aquifers which are overlaid by permeable earth material and are recharged by seeping water are called aquifers.
5. Aquifers which are sandwiched between two impermeable layers of rocks or sediments are called aquifers.
6. Shallow depressions filled with water forms and
7. Examples of wetlands are and
8. Typhoid is called by a
9. Concentration of fluoride should not exceeds mg/L in drinking water.

10. Uranium mining is done in in A.P.
11. can be extracted from bauxite ore.
12. Deforestation means of trees.
13. Forests can absorb gases.
14. Oxides of sulphur react with atmospheric moisture/rain water to form acid.
15. bacteria convert organic-N into ammonia.

(B) CHOOSE THE CORRECT ANSWER

1. During photosynthesis trees produce
 - (a) Oxygen
 - (b) Carbon dioxide
 - (c) Nitrogen
 - (d) Carbon monoxide.
2. Forests prevent soil erosion by binding soil particles in their
 - (a) Stems
 - (b) Leaves
 - (c) Roots
 - (d) Buds
3. Cholera is called by
 - (a) Bacterium
 - (b) insect
 - (c) Fungus
 - (d) Snail
4. Forests produce
 - (a) oxygen
 - (b) medicines
 - (c) Timber
 - (d) all of these.
5. Water borne disease is
 - (a) cholera
 - (b) dysentery
 - (c) typhoid
 - (d) all of these.

(C) WRITE TRUE OR FALSE

1. Surface water is more in quantity than the ground water. (True/False)
2. Mining of uranium exposes local people to radiation hazard. (True/False)
3. Soil erosion helps to maintain fertility. (True/False)
4. Optimal level of fluoride is require in drinking water. (True/False)
5. Forests are sinks for CO₂. (True/False)